

CSS 430: Operating Systems Concepts

University of Washington Bothell

Location

INV-111

Schedule

Mon & Wed
5:45 PM - 7:45 PM

Instructor

Stephen Dame

Office Hours

MW 1:00-3:00 PM
(CSS Conf Room)

Course Description

Dive deep into the architecture of modern operating systems with this comprehensive course. Using C/C++ and Java, we will explore fundamental OS concepts like processes, threads, and memory management. We will also investigate advanced topics such as synchronization, deadlocks, and virtual memory. To stay current with industry trends, we'll utilize cloud compute virtualization, software repositories and generative AI tools, providing a lite introductory foundation for modern software development as we explore the programming assignments.

Prerequisites: CSS 343

Weekly Schedule - Winter 2026

Note that this is a tentative ordering of OS topics. Chapters refer to *Operating System Concepts, 10th Ed.*

	Week Topic Focus	Session Details
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1	Intro & OS Structures	Mon Jan 5 (Ch 1) Introduction, In-class Linux orientation Wed Jan 7 (Ch 2) OS Structures, Linux Lab Setup Assignment: GCP Environment Setup Program 1 Assigned
2	Processes & Threads	Mon Jan 12 (Ch 3) Process Management Wed Jan 14 (Ch 4) Threads & Concurrency
3	Scheduling	Mon Jan 19 No Class - MLK Day Wed Jan 21 (Ch 5) CPU Scheduling Part 1 Program 1 Due Program 2 Assigned
4	Scheduling & Sync	Mon Jan 26 (Ch 5) CPU Scheduling Part 2 Wed Jan 28 (Ch 6) Synchronization Tools
5	Sync Examples	Mon Feb 2 (Ch 7) Synchronization Examples Wed Feb 4 Extra Lab Time and Q&A Program 2 Due Program 3 Assigned

6	Midterm Week	Mon Feb 9 Midterm Exam Review (Weeks 1-5) Wed Feb 11 Midterm Exam (In Class)
7	Deadlocks	Mon Feb 16 No Class - Presidents' Day Wed Feb 18 (Ch 8) Deadlocks
8	Memory Management	Mon Feb 23 (Ch 9) Main Memory - Part 1 Wed Feb 25 (Ch 9) Main Memory - Part 2 Program 3 Due Program 4 Assigned
9	Virtual Memory	Mon Mar 2 (Ch 10) Virtual Memory - Part 1 Wed Mar 4 (Ch 10) Virtual Memory - Part 2 Final Project Assigned
10	File Systems	Mon Mar 9 (Ch 13) File-System Interface Wed Mar 11 (Ch 14) File-System Implementation Program 4 Due

11	Finals Week Mon Mar 16 Final Exam Review (Weeks 7-10) Wed Mar 18 Final Exam (In Class) Final Project Due at 11:59pm
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Coursework & Grading

Coursework	Percentage	Points
Program 1	9%	90
Program 2	9%	90
Program 3	9%	90
Program 4	9%	90
Midterm Exam	25%	250
Final Exam	25%	250
Final Project	14%	140
TOTAL	100%	1000

Grading Standards

All grades will be listed as percentages in the Canvas LMS. The conversion table below will help you determine your final grade. For example, if you earn 950 points (or 95%) you will get a 4.0. If you earn 750 points (or 75%) you will earn a 2.0.

A - 100%-90%	B - 89%-80%	C - 79%-70%	D - 69%-62%	E - 61%-below
100%–95% = 4.0	89% = 3.4	79% = 2.4	69% = 1.4	= .6 (Failing)
94% = 3.9	88% = 3.3	78% = 2.3	68% = 1.3	
93% = 3.8	87% = 3.2	77% = 2.2	67% = 1.2	
92% = 3.7	86% = 3.1	76% = 2.1	66% = 1.1	
91% = 3.6	85% = 3.0	75% = 2.0	65% = 1.0	
90% = 3.5	84% = 2.9	74% = 1.9	64% = 0.9	
	83% = 2.8	73% = 1.8	63% = 0.8	
	82% = 2.7	72% = 1.7	62% = 0.7	
	81% = 2.6	71% = 1.6		
	80% = 2.5	70% = 1.5		

Total of 1000 points are baseline:

- The baseline points above are the standard allocations.
- Weekly homework assignments count as extra credit (1-2 points each depending on the difficulty of the homework assignment)
- Late assignment turn-in will result in 30% reduction in grade for each (school) day the assignment is late (e.g. Final Score = GradedScore*0.70 ^ (number of school days late)).
- Weekly quizzes count as typically 10 points each and can boost overall scores for the quarter. Quizzes must be completed on the assigned deadline in order to count.

Course Policies & Resources

Communication & Canvas

Canvas Learning Management System (LMS) will be used for notices, discussions, assignment submissions, and grade logs. Please visit the [Canvas Site](#) to access your page for this class.

Course Goals

The overall goal of CSS 430 is to learn fundamental concepts that are used in and applicable to a variety of operating systems. The course consists of three major concepts:

1. **Process Management:** Scheduling, executing, synchronizing, and terminating application programs.
2. **Memory Management:** Loading programs and dynamically allocating/deallocating memory space.

3. **File Systems:** Mechanisms for storage and access to data and programs residing on disks.

Through the course, we will use Java to illustrate many operating-system concepts. As with most technical courses, expect to spend an additional 10 to 15 hours a week outside of class time including reading, independent research, and coding.

Textbooks & References

- **Required:** [Operating System Concepts](#), 10th Ed, Silberschatz et al. (zyBook ISBN: 979-8-203-08082-0) or [UBookstore Loose-Leaf](#)
- **Java Reference:** A Java book of your choice (e.g., *Absolute Java* by Savitch or *Java How to Program* by Deitel).
- **OS References:** *Understanding the Linux Kernel* (Bovet/Cesati) or *The Design of the Unix Operating System* (Bach).

Assignment Turn-in Policy

Assignments (Programs, Labs, and Homework) are due based on the deadline specified in the Canvas LMS *Assignment* page. Please submit materials via the Canvas upload format specified. Late submissions may be accepted only in exceptional events such as accidents, sickness, sudden business trips, and family emergencies. No make-up exams will be given except under exceptional circumstances. Barring emergencies, I must be informed before the exam if you plan to be absent.

Academic Integrity

All programs and labs, except the final project, are to be performed independently. You may discuss the problem statement and any clarification with each other, but any actual work to be turned in, must be done without collaboration.

The final project may be done by a team of two, (or three students if you cannot find a single partner), in which case each student must achieve an equal amount of work. (If you work with two other students, you must explain the reasons in your report, otherwise you will receive only 2/3 of the full score.) For the detailed instructions, see the project assignment sheet in Canvas.

Special Emphasis: Use of AI

Utilization of AI is an important *new skill* that is supported in this class and by the instructor provided that plagiarism is strictly avoided. In other words, AI can be used as a tool to enhance the quality of the work produced by students but academic integrity must always be at the forefront of the product being produced and turned in as being authored by the individual student or student groups.

Access and Accommodations

Your experience in this class is important to me. It is the policy and practice of the University of Washington to create inclusive and accessible learning environments consistent with federal and state law. If you have already established accommodations with Disability Resources for Students (DRS), please activate your accommodations via myDRS so we can discuss how they will be implemented in this course.

If you have not yet established services through DRS, but have a temporary health condition or permanent disability that requires accommodations (conditions include but not limited to; mental health, attention-related, learning, vision, hearing, physical or health impacts), contact DRS directly to set up an Access Plan. DRS facilitates the interactive process that establishes reasonable accommodations. Contact DRS at disability.uw.edu.

For Our Veterans

Welcome! We at UW Bothell understand that the transition into civilian life can be challenging for our veteran students and we have many resources for any who may want to reach out for guidance or assistance. This includes our Vet Corp Navigator through the WDVA and our Student Veterans Association (SVA). Please contact Veteran Services at 425.352.5307 or rosal@uw.edu. For those of you needing more URGENT support, please call The Suicide Prevention Hotline 1.800.273.8255 or connect with the UWB CARE Team <https://www.uwb.edu/studentaffairs/care-team>.